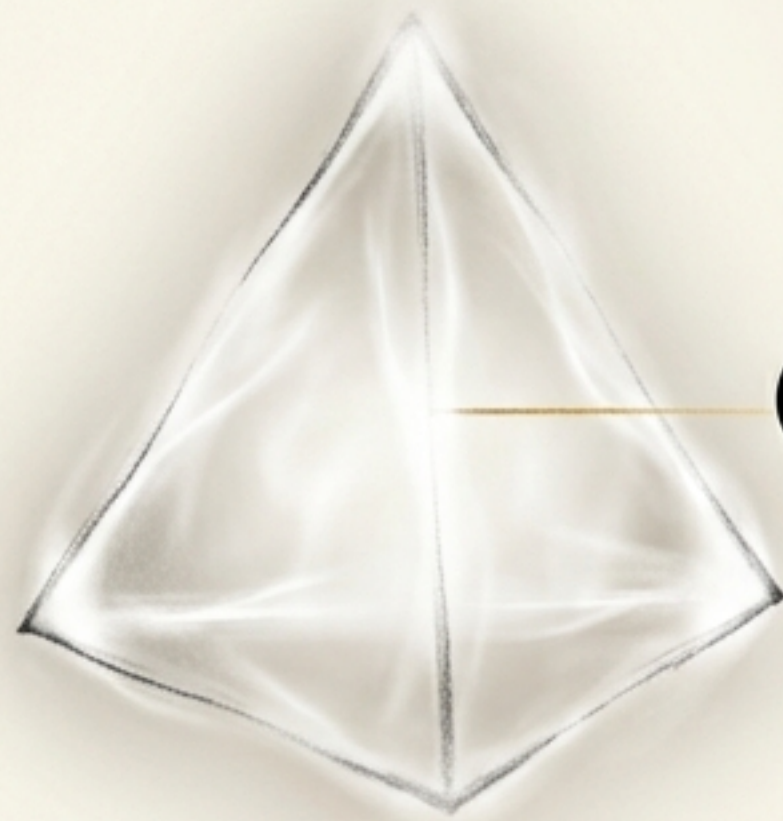
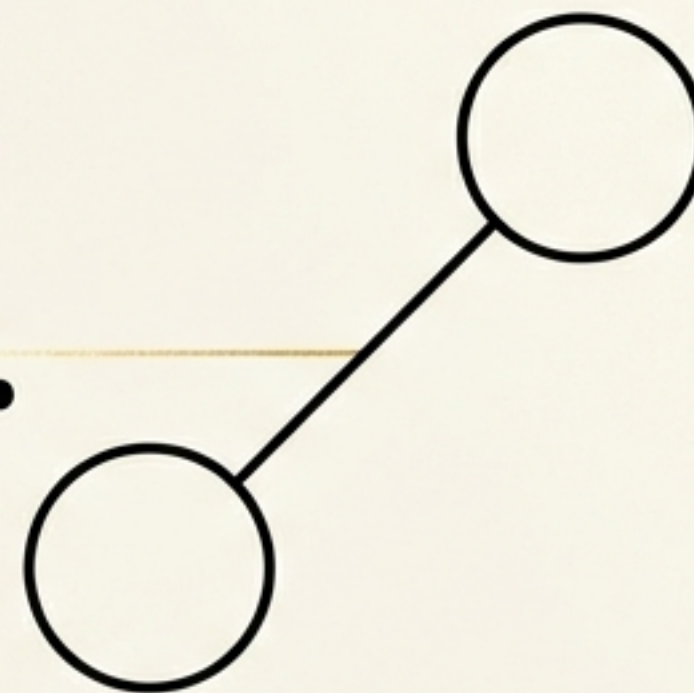
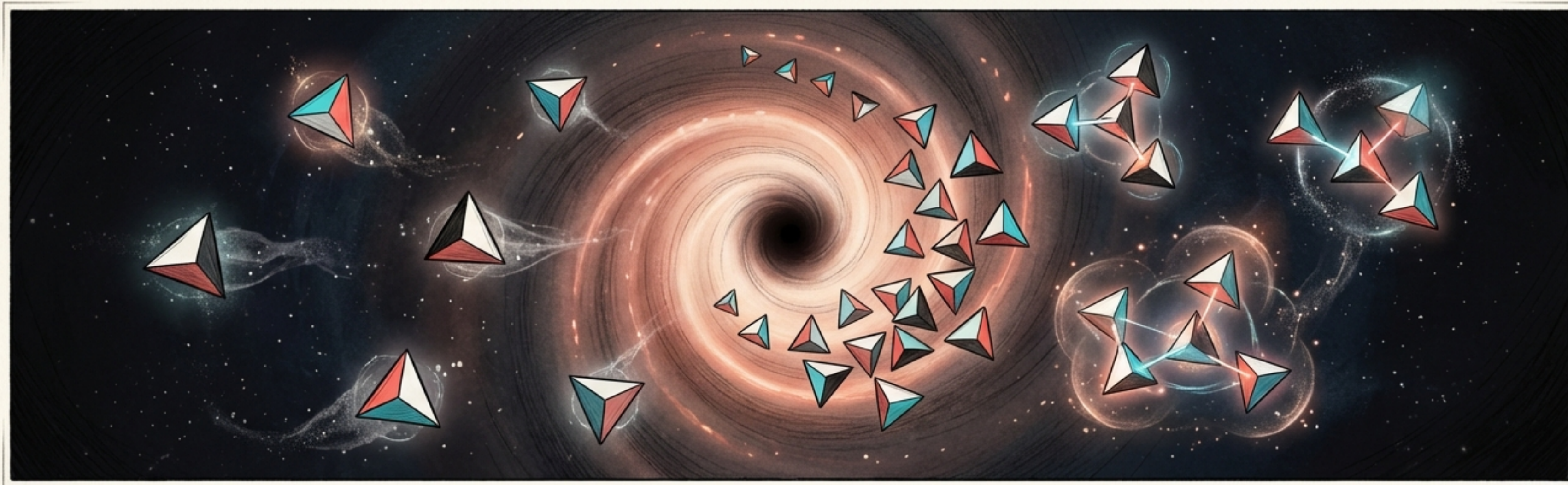




Geometry is Destiny.



TET~CRAFT: An Emergent Universe of Fact and Form



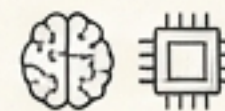
Key Descriptor

A kleiniverse of tetrahedrons that bond into atoms, react, and evolve into complex molecules.



Core Concepts

- **The Simulation:** An emergent quantum chemistry and metaphysics simulation (v5.3D).
- **The Building Block:** Each tetrahedron (TET) represents a quantum fact that can combine with others to form understanding.
- **The Goal:** To simulate the formation of understanding from disconnected facts, exploring quantum bonding as a computational metaphor.



TET
Fact

Fact



Molecule
Understanding

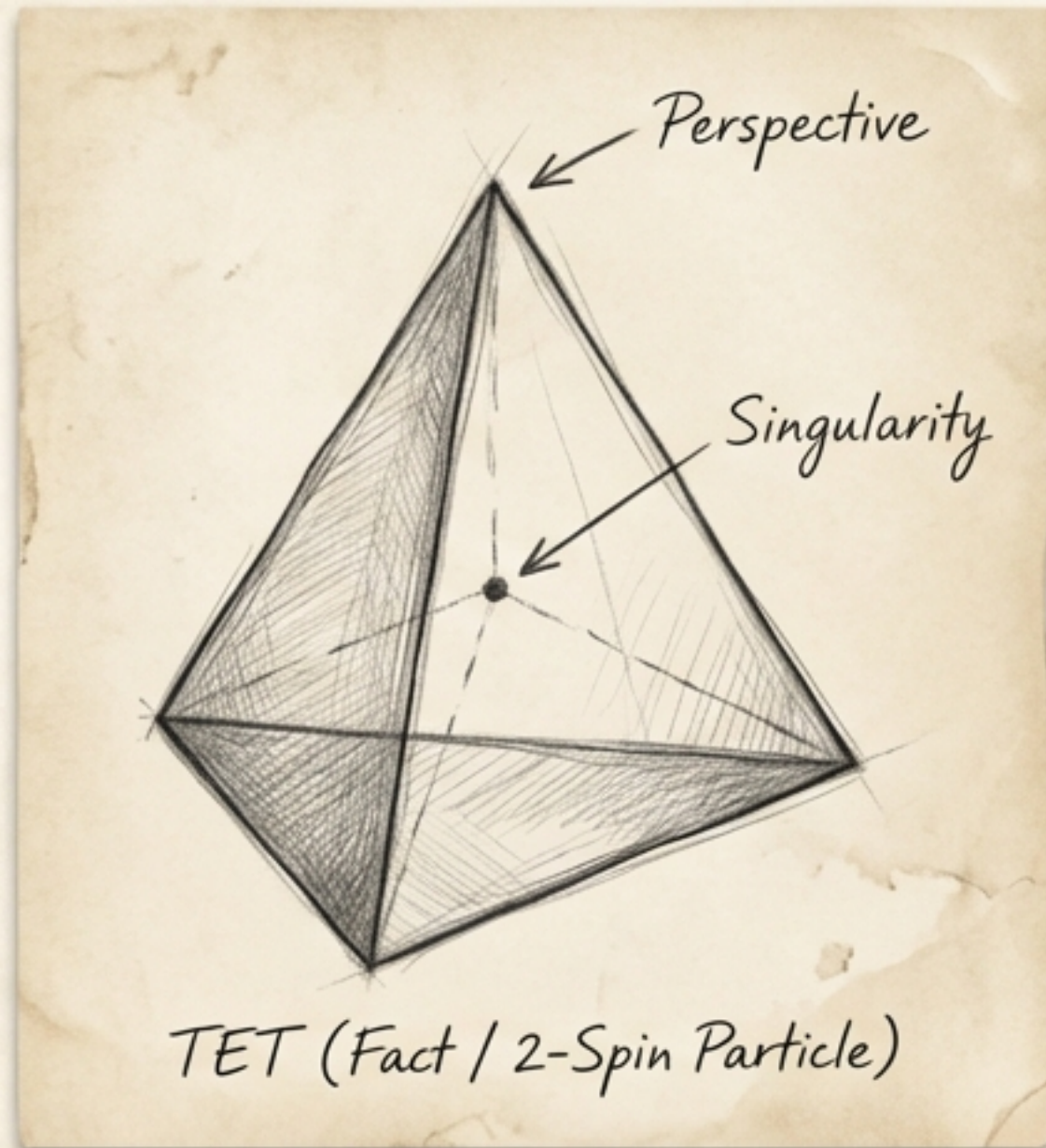
Evolution



The Architect's Vision: The Metaphysical Blueprint

The project's "vibe" is rooted in a speculative, metaphysical framework where geometry and consciousness are intertwined.

"God is the fractal to which we are each a frame."



The Player as God

The simulation assumes the player is the prime observer missing from quantum theory. Pressing SPACE creates a new (TET) to be (mis)understood.

The TET as Fact

TETs represent both facts and 2-spin particles, the most basic 3D structure. Connecting them slowly coheres them into (mis)understanding.

Quantum Entanglement as Desire

"Attraction/Desire" is explicitly equated with "spooky action at a distance."

Time as Misunderstanding

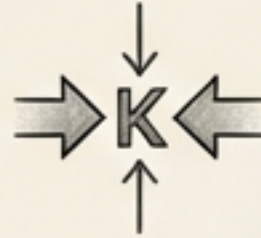
Time emerges from different perspectives and memories, allowing for discovery instead of immediate recognition.

The Forces of Creation are Mapped to the Sefirot

The simulation's physics constants are symbolic representations of divine energies, creating a dynamic Tree of Life.

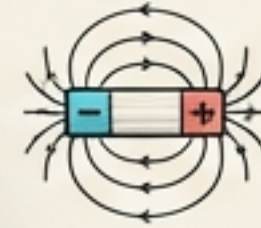
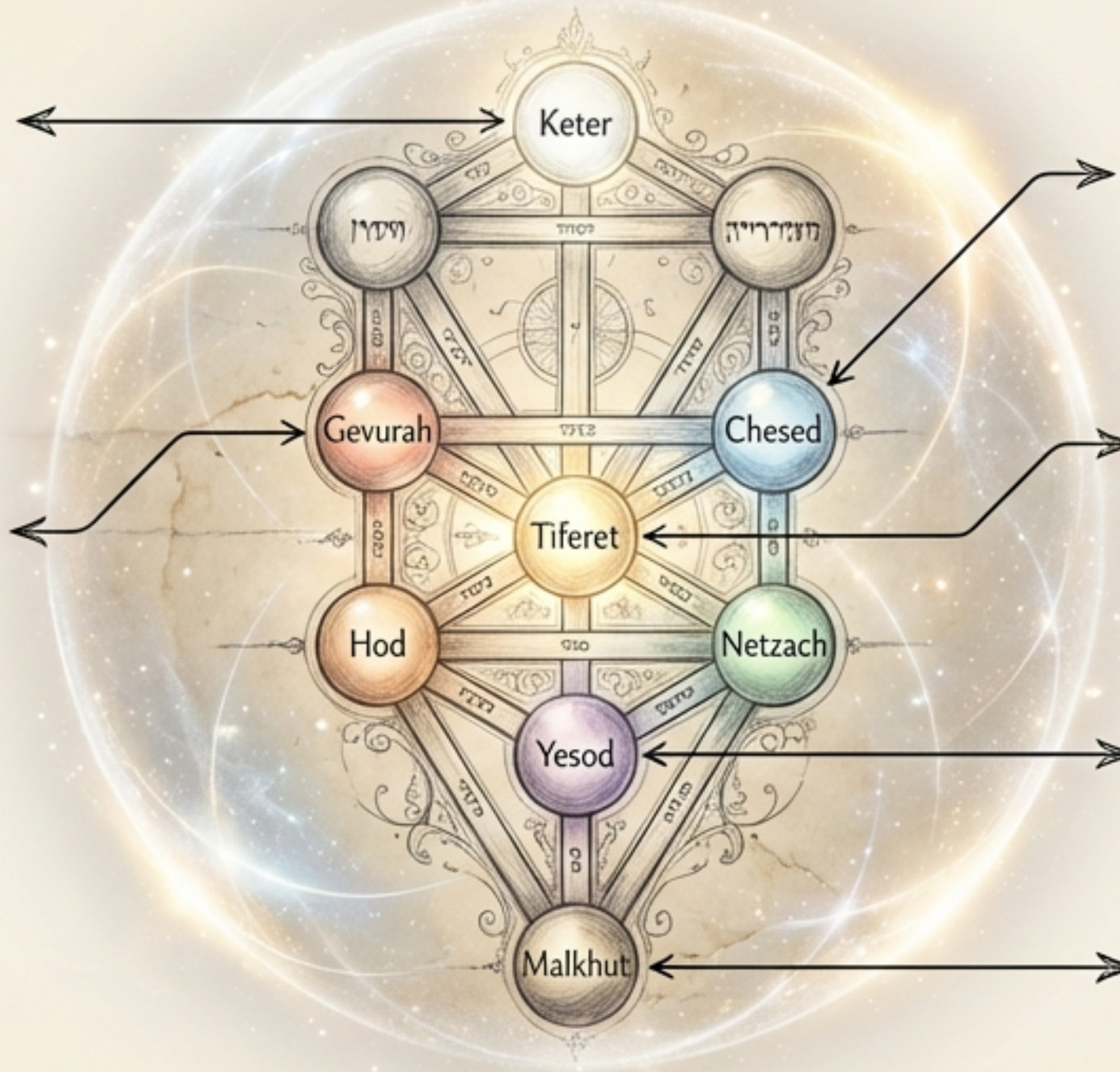
K_UNIFIED_FORCE:

Keter to Malkhut
(God's will structuring existence).



K_STICKY_PULL:

Chesed to Gevurah
(The dance between Mercy and Judgment).



Magnetism: Netzach to Hod
(The oscillation between active and passive energies).



Corner Desires: Tiferet
(The harmonious balance point reconciling opposites).



Chemical Reactions: Yesod
(Where potential forms actualize and memory accumulates).



The TETs Themselves: Malkhut
(A limited expression of the divine pattern).

The "4D Past Projection" in the simulation represents **Ain Sof** (the Infinite), the boundless light from which all forms emerge.

Color as a Channel for Divine Influence

The colors are not literal, but symbolic representations of spiritual forces. The spiritual root of every color seen in the physical world originates in its corresponding Sefirah.



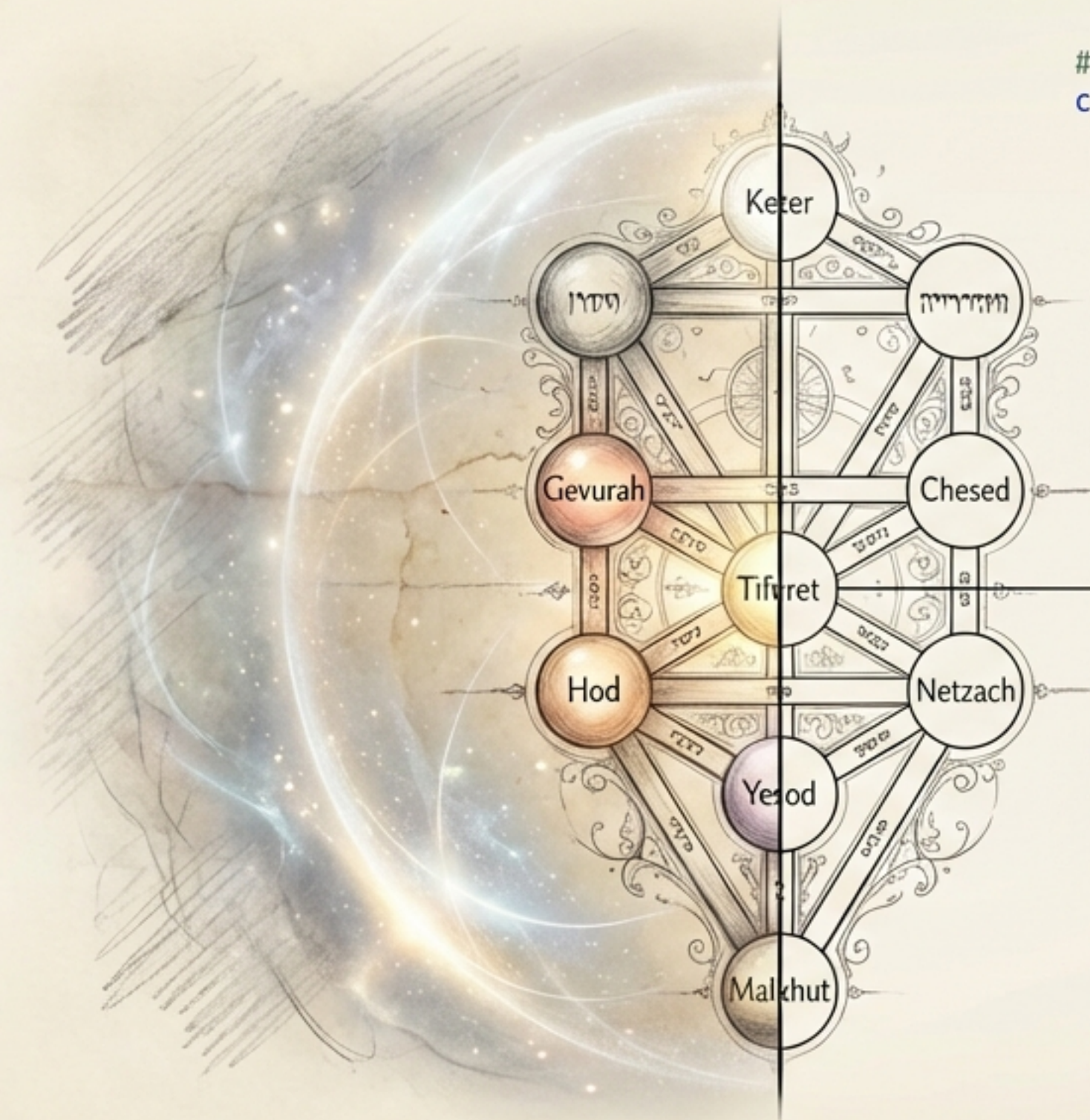
White: Corresponds to *Chesed* (Kindness/Mercy). It symbolizes peace, compassion, and benevolence. The Archangel Michael governs this line.



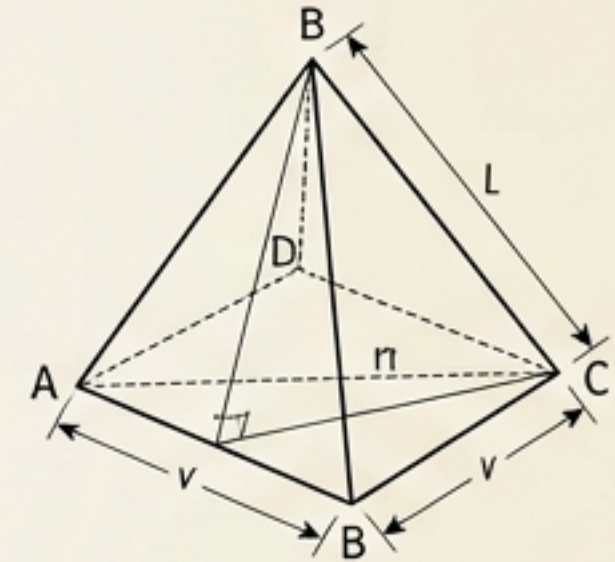
Red: Corresponds to *Gevurah* (Strength/Judgment). It signifies intensity, power, and severity. The Archangel Gabriel rules this line.

Application in the Temple: The High Priest's white garments on Yom Kippur embodied mercy and atonement. The miracle of the crimson thread turning white signaled the transformation of judgment into mercy.

From the Metaphysical to the Mechanical: The Machine's Reality



```
# Define the tetrahedron's geometry and physics
class Tetrahedron:
    def __init__(self, vertices, density=1.0):
        self.vertices = np.array(vertices)
        self.density = density
        self.mass = self.calculate_mass()
        self.inertia_tensor = self.calculate_inertia()
```



Property	Value	Units
Edge Length	1.414	m
Volume	0.2357	m ³
Mass	2.357	kg
Center of Mass	[0, 0, 0.35]	(x, y, z)

The visionary blueprint is realized through a rigid, uncompromising system of code and mathematics. Here, the laws of the physics engine are absolute, and geometry is implemented with mathematical precision.

The Core Systems are Defined by Invariants and Phases

Invariants (These **MUST** not be violated):

1. Pairing defines elements.
2. Physics $\neq$ Chemistry $\neq$ Gameplay ... yet.
3. Catalysts cannot transmute without mediators.
4. Low energy causes positional drift, not destructive apoptosis.

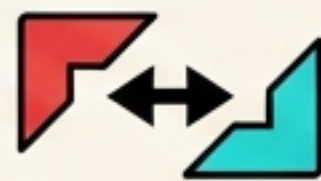
The 4-Phase Chemistry Engine



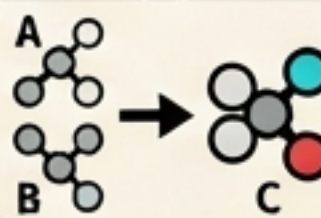
PHASE 1: Multi-Face Locking
TETs lock faces; color patterns identify molecules.



PHASE 2: Battery Oscillation
Magnetic pairs exchange "emptiness" (1 - battery).



PHASE 3: Corner Desires
Red corners seek Cyan corners; polarity rules attraction/repulsion.



PHASE 4: Chemical Reactions
Synthesis, decomposition, and catalysis govern molecular change.

Progress is Measured on The Cosmic Ladder

Core Mechanic: The universe gains Entropy/Progress Points through Synthesis Reactions or the formation of stable Geometric Interfaces.

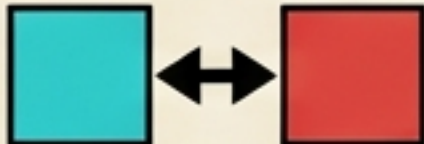
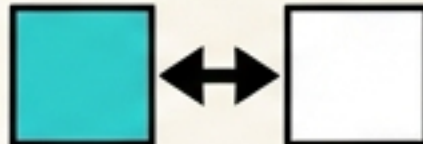
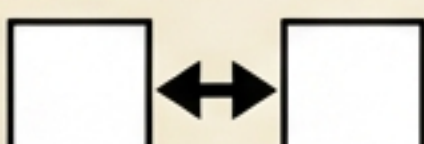
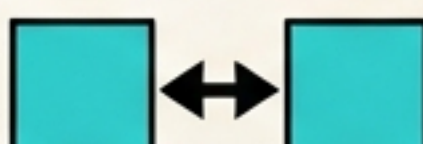
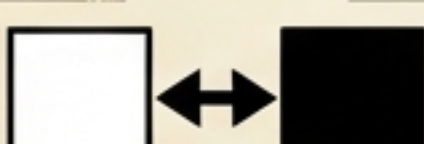
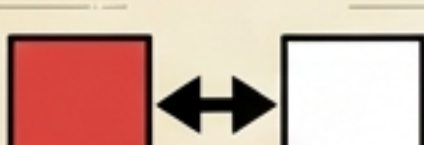
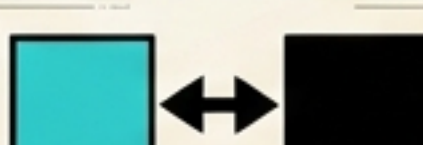
Era	Threshold	Description
1. Void	0 pts	“Nothing yet exists”
2. Mind	3 TETs	“Awareness flickers”
3. Fluctuation	3 Atoms + 0.1 PSI Req.	“Quantum foam emerges”
4. Condensation	7 pts + 0.3 Phi Req.	“Geometry aligns”
5. Chemistry	70 pts + 7 Molecules + 24 TETs	“Elements Transmute”
6. Life	777 pts	“Metabolism emerges”
7. Transcendence	[Threshold TBD – see Divergence]	“Beyond matter”

You cannot “buy” upgrades. You cannot “cheat” by renaming things.

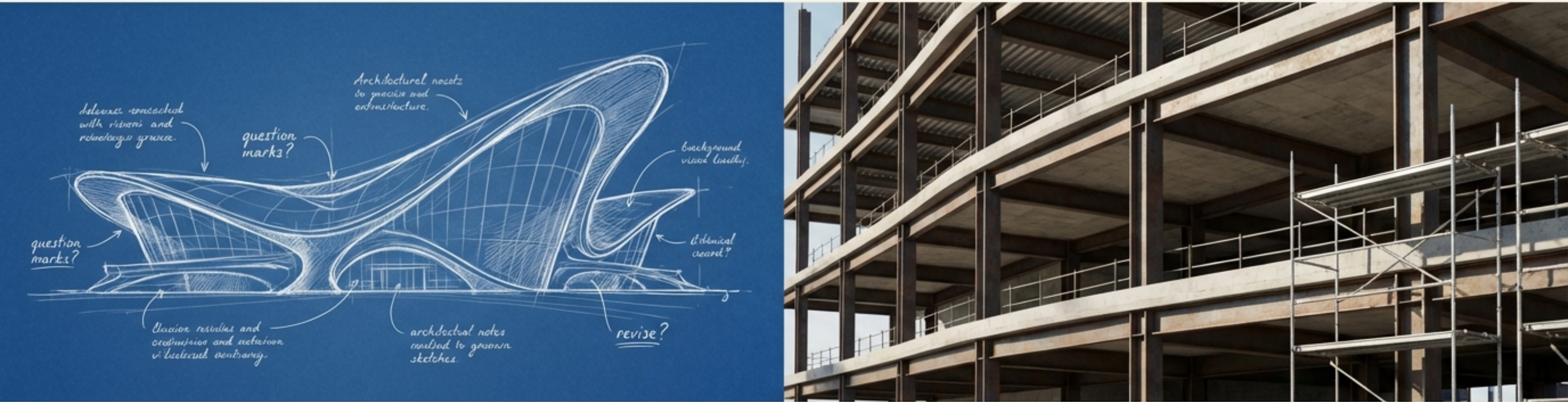
The Immutable Law of the Machine: Interface Chemistry

Elements are defined by *which* faces touch, not by arbitrary labels. You must physically connect specific faces to transmute matter.

The Interface Chemistry Table

	H Hydrogen		F Fluorine
	He Helium		Fe Iron
	C Carbon		Al Aluminium
	N Nitrogen		Cu Copper
	O Oxygen		Ce Cerium

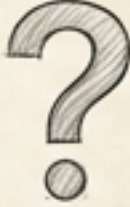
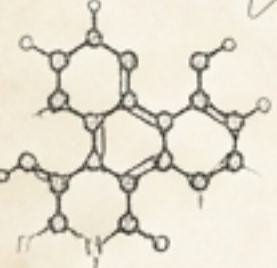
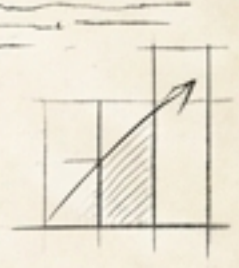
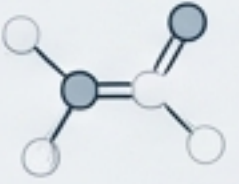
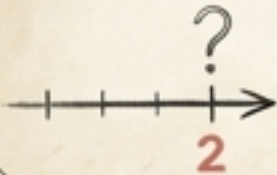
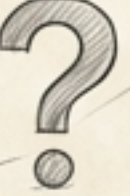
Systemic Divergence: Where the Blueprint Meets Reality



“Imagine a **blueprint versus a finished building**. The metaphysical notes are the **architect’s inspired sketches**, sometimes containing “question marks”... The Python code is the **construction supervisor’s reality**, where the building must follow the rigid laws of gravity and geometry to stand.”

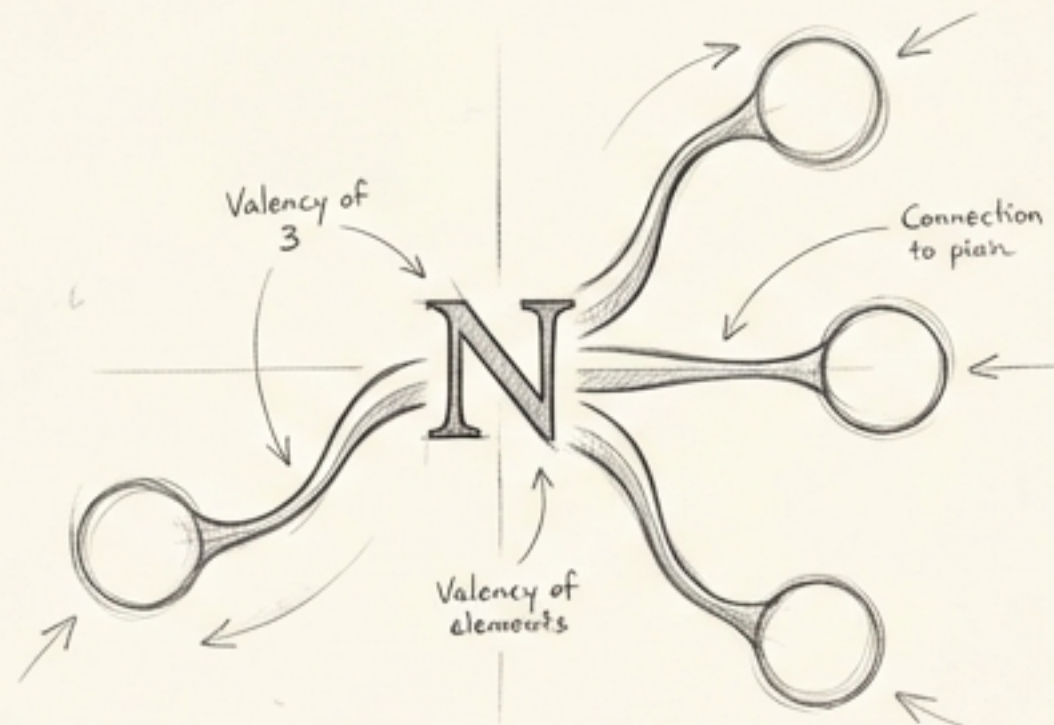
The project contains several documented discrepancies between the visionary notes and the hard-coded implementation. These divergences are not bugs, but evidence of a living dialogue between a grand vision and its practical execution.

The Dialogue Between Vision and Code Made Concrete

Concept	The Metaphysical Blueprint (Notes)	The Machine Reality (Python Code)
Transcendence Threshold	 200 Progress Points (Cosmic Ladder Table)	 777 Progress Points (<code>`TechTree`</code> class) 
Tetrahedron Geometry	 "9? edges" (Speculative note) 	 6 Edges (Defined in <code>`EDGES_NP`</code> array) 
Nitrogen's Function	 Valency of 3 (Supports structural complexity) 	 Bond Capacity of 1 (Falls under <code>`default`</code> in <code>`BOND_CAPACITY`</code>) 
Era of Fluctuation	 User query suggested it was Era 2. 	 Defined as Era 3 (Era 2 is "Mind"). 

Case Study in Divergence: The Role of Nitrogen

The Architect's Vision



“Nitrogen (N) -3 - Structure”

In the ‘Chemical Kabbalah,’ Nitrogen is described as having a valency of 3, allowing it to support structural complexity in molecules.

The Supervisor's Reality

```
BOND_CAPACITY = { #max
    'C': 4,
    'Ce': 4,
    'Fe': 2,
    'default': 1
}
```

Nitrogen is not explicitly listed in the `BOND\_CAPACITY` dictionary. Therefore, it falls under the `default` capacity of 1, restricting it to a single bond in the current implementation, despite the metaphysical design.

The Machine's Purpose: Turning Collision into Story

The simulation includes an AI layer that automates exploration and documentation.



Functions

- The Bot spawns new TETs, generates 'chemical thoughts,' and tracks synthesis champions.

The Synthesis: Emergence of Meaning from Simple Rules

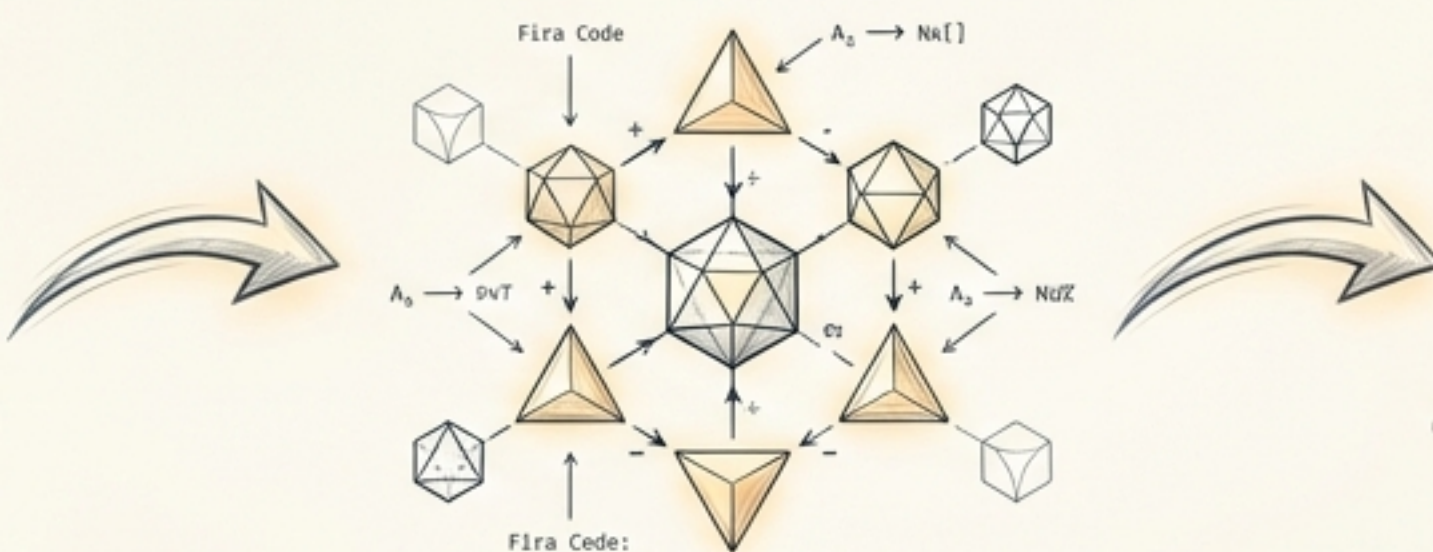
The Unified Goal

The project's end goal is the synthesis of its two pillars:

- **(From The Machine):** To achieve emergent chemistry from simple geometric rules.
- **(From The Metaphysics):** To simulate how understanding forms from disconnected facts.



1. Introduction of Facts



2. Enforcement of Laws



3. Documentation of Story

The Process & Conclusion

1. The player/creator introduces **Facts** (TETs) into the void.
2. The **Machine** enforces rigid, geometric laws, causing interactions and chemical reactions.
3. The **Bot Intelligence** observes these interactions, documenting them as a 'story.'

Conclusion: The system is designed to transform a universe of chaotic, disconnected data points into a coherent, observed, and documented narrative of understanding.



The simulation becomes a dynamic **Tree of Life**. The player (Keter) peers through the camera to witness **Malkhut (TETs)** ascend the tree through chemical reactions, all drawn toward the central **singularity** which is simultaneously the source and the destination—the alpha and omega collapsing into one another.